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PALEOMAGNETIC EVIDENCE FOR THE RELATIVE POSITIONS OF TIMOR AND AUSTRALIA IN THE PERMIAN

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An Upper Permian paleomagnetic pole has been determined for the Cribas Formation in eastern Timor. The coordinates for the mean pole are 159.8° E and 56.6° S, α_{95} = 9.0. The reliability of the pole is ascertained through thermal demagnetization, a fold test, comparison between red beds and a lava flow, and the presence of normal and reversed polarities. The Timor pole is in excellent agreement with the Australian Upper Permian and Triassic poles. From this it is inferred that autochthonous Timor formed part of the Australian continental margin at least since the Upper Permian.

1. Introduction

The Banda arc system, northwest of Australia (Fig. 1), consists of two subparallel chains of islands of which the northerly one is a typical volcanic island arc and the southerly a non-volcanic outer arc [1]. It is commonly argued that the Banda arc represents the collision zone of the Australian continent with a volcanic island arc/subduction zone. The arc provides, therefore, a unique opportunity to study the dynamic and tectonic consequences of such a collision. Timor is the largest non-volcanic island of the outer arc and its geology better known than most, largely because its complex imbricate and overthrust structures have been a subject of geological debate for more than half a century. Unravelling the tectonic history of Timor is likely to provide the key to the interpretation of the Banda arc system as a whole.

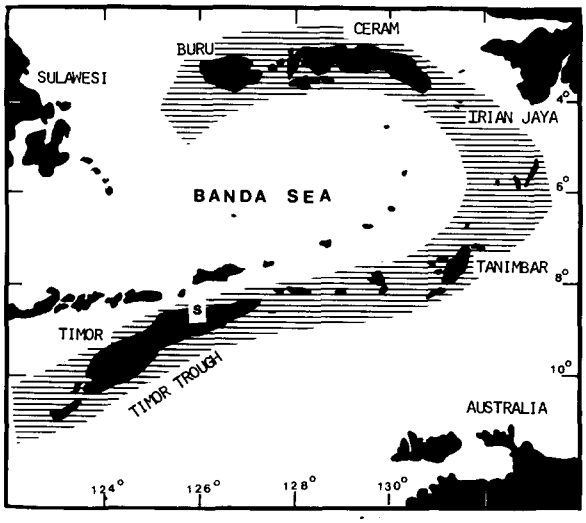


Fig. 1. Showing the Banda arc system (striped ornament) and location of Timor. Location of Fig. 2 is shown as S.

An important and unresolved issue is the relative position of Timor, Australia and the presumed subduction zone. On the one hand it has been argued [2–4] that the Timor trough should be considered as the easterly extension of the Java trench, in which case Timor, lying north of the subduction zone, is regarded as a tectonic mélange [4], produced by the advancing arc, bulldozing the shelf strata and incorporating them into the imbricated and mélanged material riding at the front of the arc [4]. In support of this hypothesis parallels are drawn between Timor and the outer arc islands and ridges, southwest of Sumatra and Java and

the marine seismic records of the Timor trough [4,5] interpreted to show the sediments are dipping northward under Timor. The alternative hypothesis, originally proposed by Teichert [6], and elaborated more recently by Audley-Charles and others [7,8], assumes that at least part of Timor (para-autochthonous Timor) has been part of the Australian continental margin since the Permian. This hypothesis is based on the stratigraphical and structural interpretation of some Permian and Mesozoic rocks, and in particular the faunal affinities they exhibit with Australia. To explain the complicated tectonic structures and the origin of allochthonous Timor, a speculative paleographical reconstruction has been proposed [8] in which outer arc islands are supposed to have existed south of Java in the Permian which were subsequently thrusted horizontally over Timor in post-Miocene times.

The latter hypothesis implies that there existed crustal continuity between Timor and Australia, and it places the subduction zone to the north of Timor. Chamalaun et al. [9] have shown that the gravity field of eastern Timor is consistent with the concept of an Australian continental crust being continuous and extending towards the Banda Sea, as far as the north coast of Timor. In this paper an alternative test of the continuity hypothesis is reported. It is based on the proposition that if the hypothesis is correct the Permian paleomagnetic poles of autochthonous Timor should coincide with the Australian Permian poles. Non-coincidence could be due to a variety of reasons. In the present context, for example, one would not expect the Australian strata to have preserved their original attitudes, if they were incorporated into the mélange when it ramped into the continental shelf. The direction of magnetization of the formations involved would be randomly orientated and the corresponding poles significantly different from the Australian ones. Similarly if the relative positions of Timor and Australia had changed significantly since the Permian this would be expressed by differences in the pole positions.

2. Cribas Formation

In his description of the geology of Timor, Audley-Charles [10] designates twenty-four formations, of which four are considered to be wholly allochthonous. The autochthonous formations (more recently termed para-autochthonous [8]) range in age from the Lower Permian to Recent. The oldest of these is the Lower Permian Atahoc Formation, whose type section is the Cribas anticline south of Manatuto (see Fig. 2). It is overlain conformably by the Cribas Formation, which contains several hundred meters of red beds, and therefore, appeared to be an encouraging paleomagnetic target to perform the test described above.

The Cribas Formation is mapped [10] as a broad anticline (Cribas anticline) with a fold axis running roughly east-west. It appears [10] to plunge west below one klippe of the Lolotoi complex and east below another. In the sample area, bedding plane attitude varies considerably and the structures are more complicated than implied above. The age of folding is not firmly established as it depends critically on the interpretation of the overthrust history of Timor. However, Audley-Charles [10] suggests that the pre-Eocene rocks were folded between the Senonian and Middle Eocene and more recently revised his estimate to suggest a Pleistocene age [8].

The formation was assigned an Upper Permian age on the basis of relatively sparse fauna [10]. In the type section the overlying Triassic Aituto Formation (Ladinian) is not seen because there is no clear lithological break. However, the upper boundary of the Cribas Formation is taken to be the first horizon containing the Triassic index fossil, *Halobia.* In the type section the Cribas Formation is composed of shales, micaceous, red and green shales, red arenites and red calcareous mudstones and includes rare lava flows.

3. Paleomagnetism

The paleomagnetic collection was undertaken in 1974 and limited to the type section in the river Sumasse just west of the village of Cribas. Some 60 samples were obtained throughout the exposure, but the rocks are extremely fissile and prone to breakage during transport and drilling. In fact, half the collection was lost and often only one cylindrical specimen per sample was retrieved. Further losses occurred during thermal demagnetization, when specimens exploded at about 400°C. The surviving samples cover approximately 200 m of red beds, and are spaced more or less

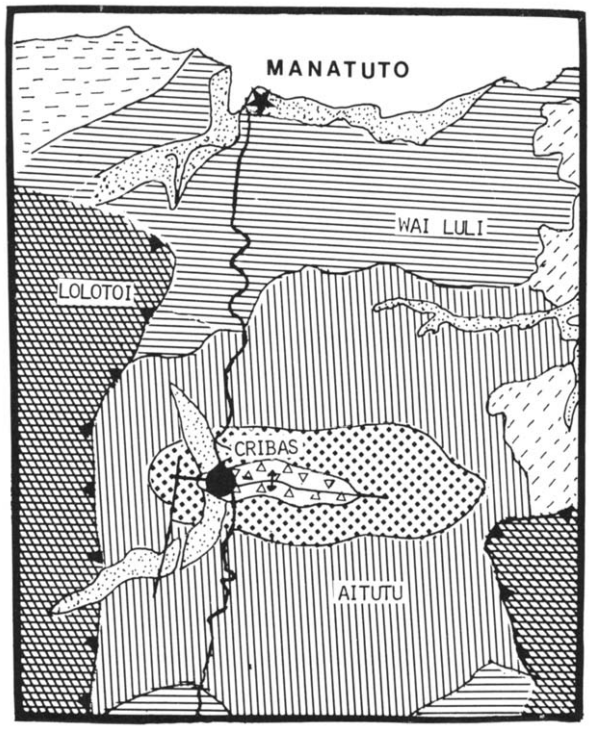


Fig. 2. Showing sampling location in the Cribas anticline, with surrounding geology as marked. Lolotoi: pre-Permian metamorphic complex, Aitutu: Triassic and Wai Luli: Jurassic. Redrawn after Audley-Charles [10].

uniformly throughout this thickness. Only one sample was collected from each recognizable horizon, but some duplication of horizons could have occurred inadvertedly as a result of faulting in the area. As the actual time span covered by the collection is not known, and no clear lithological breaks observed which might suggest separate time zones, the collection has not been divided into horizons or sites. The results described below are therefore, based on giving unit weight to sample means. Stratigraphically the collection starts at the Atahoc Formation and extends upwards into the exposed Cribas Formation. However, the black silty shales at the base of the section were not sampled as they were judged paleomagnetically unsuitable. The sampling locality and surrounding geology is shown in Fig. 2.

The NRM directions are shown with respect to the horizontal and the bedding plane in Fig. 3A and B. Both figures show a considerable scatter in the directions. Nevertheless, with respect to the bedding planes the majority of samples possess a direction of magnetization well away from the present field, indicating the presence of a stable remanence. The scatter is suggestive of variable secondary components and consequently all samples were subjected to progressive thermal demagnetization to 625°C. While a few specimens reached a stable end point at 500°C, most required thermal demagnetization to 625°C. The resultant cleaned directions are shown with respect to the horizontal and bedding planes in Fig. 3C and D. With respect to the bedding plane, the cleaned directions are tightly grouped and produce a well-defined mean direction with high positive inclination. The statistical details are set out in Table 1.

A few samples are found with normal polarities. They are too few in numbers to calculate a separate mean, and their polarity has been artificially reversed so as to include them in the mean calculations set out in Table 1.

The collection included one amygdaloidal lava.

TABLE 1

Mean directions and statistical parameters for the Cribas Formation (site co-ordinates 126°E, 9.0°S)

	N	*R*	*K*	*D*	*I*
Before thermal demagnetization; with respect to horizontal	32	14.098	1.73	136.2	8.3
Before thermal demagnetization; with respect to bedding	32	14.230	1.74	149.3	35.7
After thermal demagnetization; with respect to horizontal	32	27.899	7.56	131.1	33.1
After thermal demagnetization; with respect to bedding	32	29.743	13.73	158.6	53.3

N = number of samples, *R* = resultant vector, *K* = precision estimate, *D* = declination (degrees), *I* = inclination (degrees).

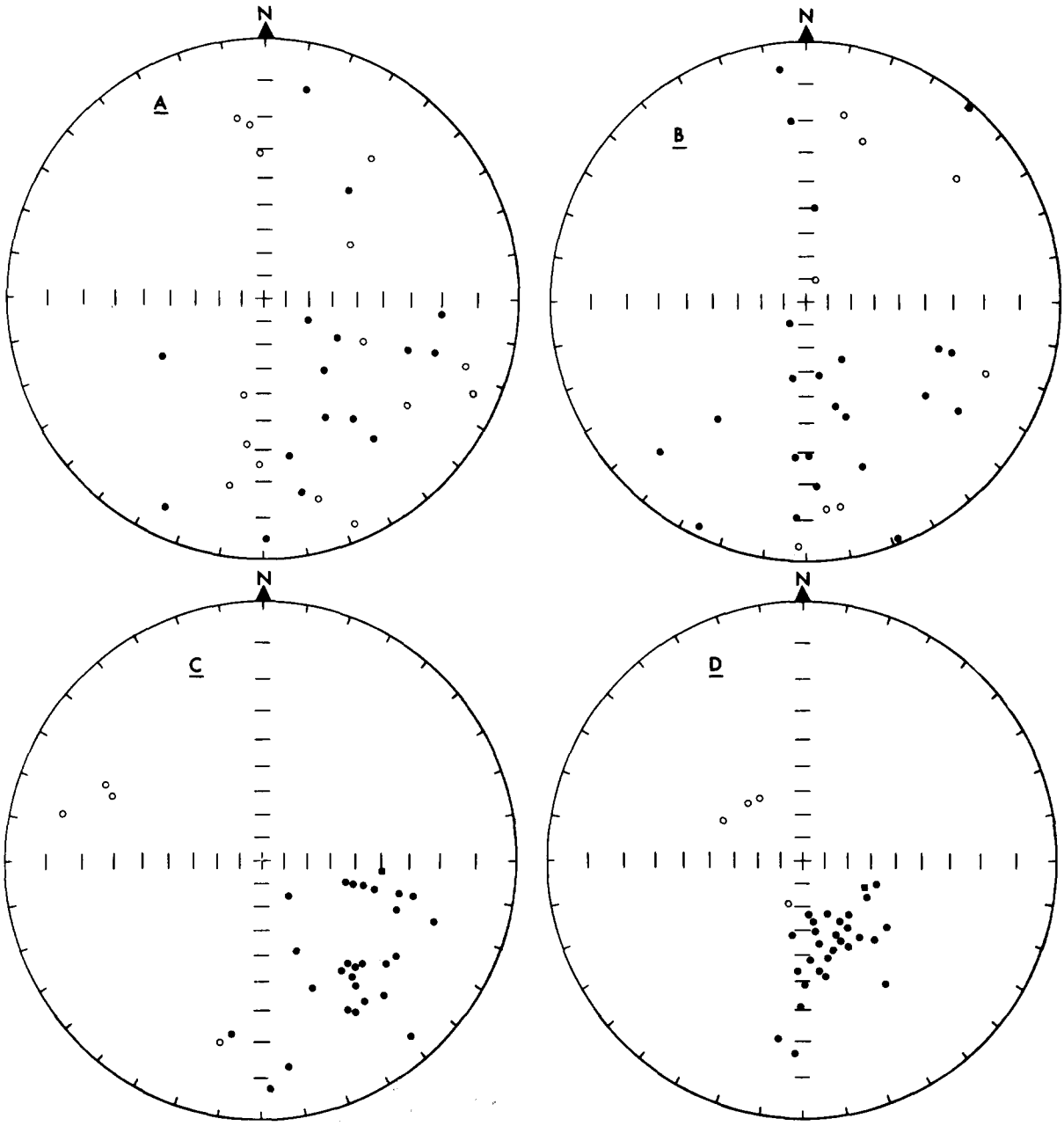


Fig. 3. Showing the directions of magnetization: (A) before thermal demagnetization with respect to present horizontal; (B) before thermal demagnetization with respect to bedding plane; (C) after thermal demagnetization with respect to horizontal; (D) after thermal demagnetization with respect to bedding plane. Amygdaloidal lava is shown with square symbol.

Its direction is shown by a square symbol. The basalt occurs in an isolated outcrop and its attitude correction is based on the bedding planes of the adjacent sediments. The basalt direction agrees with that of the sediments.

The figures show a decrease in the dispersion when

the directions are corrected for folding. The statistical test for the significance of the fold test [11] is significant at the 95% level, although the surviving samples are mainly from the southerly limb of the fold. Examining the change in direction of individual samples with widely divergent bedding planes confirms that the magnetization was clearly acquired before folding.

The Cribas Formation yields a mean direction after thermal cleaning of $D = 158.6°$, $I = +53.5°$ with $\alpha_{95} = 7.1°$. This gives a pole at 58.3°S, 161.3°E with $dm = 6.7°$ and $dp = 9.6°$. The corresponding mean of the poles has co-ordinates 56.6°S, 159.8°E and $\alpha_{95} = 9.0$ The reliability is firmly established by the fold test, stability of magnetization during thermal demagnetization, the presence of normal and reversed polarities, agreement between the lava flow and red beds.

4. Discussion

In Fig. 4 are shown the Permo-Triassic poles for Australia together with their error circles, and the Upper Permian Cribas Formation for eastern Timor. The Australian poles have been numbered in order of decreasing age. Poles 1 and 2 correspond to Lower Marine and Upper Marine [12], which are Lower and Upper Permian in age respectively. K/Ar dating of the Upper Marine Latite gave an age of 248 m.y. [12]. The Milton Monzonite (pole 3 in Fig. 4), was dated at 240 m.y. Its pole position relative to the Upper Marine Latite appears anomalous. However, it was suggested [13] that it may have to be subjected to a tectonic correction of 15°, which would bring it in closer agreement with the Upper Marine Latite. The Australian poles 4 and 5 correspond to the Narrabeen chocolate shales [14] and Brisbane Tuff [15], which are Lower and Middle Triassic, respectively, on stratigraphical grounds. The Cribas pole is statistically indistinguishable from either of these.

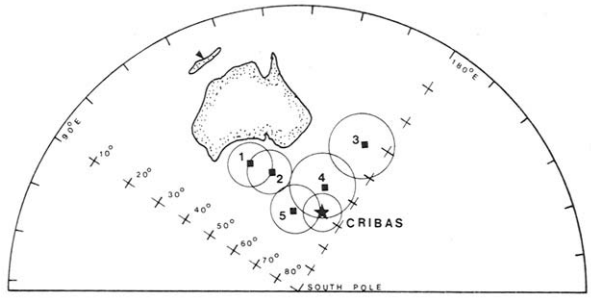


Fig. 4. Showing Australian Permo-Triassic poles and the Cribas pole. 1 = Lower Marine (Lower Permian) [12], 2 = Upper Marine (Upper Permian, 248 m.y.) [12], 3 = Milton Monzonite [14], 4 = Narrabeen chocolate shales (Lower Triassic) [16] and 5 = Brisbane Tuff (Middle Triassic) [15].

The agreement between the Cribas pole and the Australian Permo-Triassic poles, lends strong support to the hypothesis that the Cribas Formation and, therefore, para-autochthonous Timor was part of the Australian continental margin. On closer examination it may be argued that, nevertheless, the Cribas pole is slightly too far to the east, and should, on account of its age, be situated between the Australian poles of Upper Marine Latite and the Lower Triassic Narrabeen group. The difference is about 12° which is not regarded as significant in view of the uncertainties in stratigraphical correlation and experimental errors. For example, no correction was made for plunge of the fold, which is thought to be low. Furthermore, there is much evidence for post-Eocene tectonism in Timor [10], which may have resulted in a small tectonic tilt or, alternatively, it may be speculated that some crustal shortening, perhaps with the attendant formation of the Timor trough, occurred as a result of the collision. Furthermore, it may be noted that Schmidt [16] presented recently some new Mesozoic data for Australia, which indicated a Jurassic pole further to the east than previously thought and in good agreement with equivalent poles from other parts of Gondwanaland using the reconstructions of Sproll and Dietz [17] and Smith and Hallam [18]. While on present indications the Cribas Formation is older, i.e., Upper Permian rather than Jurassic, a more easterly polar shift since the Permian in the Australian polar wander curve, could perhaps also explain the easterly position of the Cribas pole. The relative importance of these effects can only be evaluated when a detailed polar wander curve for Timor becomes available for comparison with Australia and a reconstructed Gondwanaland.

Finally, it is interesting to note that Audley-Charles [8] suggested that the Cribas Formation was formed in a pro-deltaic environment, with the detrital material derived under rigorous climatic conditions. The magnetic inclination places Timor in 34° which,

while not providing strong support for Audley-Charles' [8] observation, is not totally inconsistent. On the other hand, it is consistent with the occurrence of abundant red beds.

5. Conclusions

It is concluded that autochthonous Timor formed part of the Australian continent since the Permian, and that the paleomagnetic data presented here do not lend support for the mélange theory of Fitch and Hamilton [4].

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